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Towards Effective Governance in Energy Public–Private Partnership Projects: Evidence from Myanmar's Developing Economy

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Abstract: Evidence on energy-sector Public–Private Partnership (PPP) governance in developing economies remains sparse and often extrapolated from non-sectoral or developed-country studies, limiting guidance for union-level decision makers. This paper addresses that gap by providing case-verified evidence and a replicable governance framework from Myanmar's experience. Using a qualitative comparative multiple-case design, we examine four union-level energy PPPs—two hydropower and two oil-and-gas, including two successful and two unsuccessful—via semi-structured interviews and document review, with cross-case synthesis. We assess legal–regulatory arrangements, institutional capacity and coordination, risk allocation and financial structuring, stakeholder engagement, and policy alignment. The study identifies 22 critical governance factors associated with performance, notably legal compliance, transparent procurement, effective regulatory oversight, and fit-for-purpose financial mechanisms. Constraints include limited institutional resources, weak protection of contractual and intellectual-property rights, and policy–practice misalignment. Three actionable insights emerge: centralized oversight must be paired with inter-agency coordination; policy–practice gaps undermine implementation; and financial innovation with clear risk communication improves project viability. We contribute a practice-tested, transferable framework linking governance mechanisms to energy PPP performance in resource-constrained settings, informing policy design, contract management, and capacity building.

Keywords: Energy infrastructure; Governance frameworks; Hydropower projects; Myanmar case study; Oil and gas projects; Public–Private Partnerships

1 Introduction

Public–Private Partnerships (PPPs) are widely used to bridge infrastructure gaps in developing economies by combining private-sector innovation with public accountability [1]. PPP models—Build-Operate-Transfer, Design-Build-Finance-Operate, and concessions—structure risk-sharing, financing, and ownership arrangements [2]. In the energy sector, Build-Operate-Transfer typically serves capital-intensive generation assets, while Design-Build-Finance-Operate is suited to transmission; the choice of model depends on asset scale, risk allocation, and contract duration [3, 4] . Given the long-lived and complex nature of PPP contracts, governance arrangements are central to transparency, stakeholder trust, and operational performance [5].

Myanmar's energy mix relies primarily on hydropower and gas, yet persistent supply gaps and low rural electrification rates persist [6]. PPPs are applied to large grid-connected assets, whereas microgrids and off-grid solutions face high transaction costs, lower returns, and regulatory hurdles that constrain PPP suitability [7]. On the finance side, foreign direct investment has backed major hydropower and gas ventures (e.g., Upper Yeywa, Myitsone), often through joint ventures, but regulatory and environmental concerns have complicated their implementation [8]. Official Development Assistance—for instance, the National Electrification Project—has expanded access without necessarily relying on private participation [9, 10]. Remittances also play a complementary role by financing household-level solar and microgrids, though such investments typically lack durable institutional or regulatory support [9]. While such programs are vital for rural development, they can alter private risk perceptions and incentives, shaping the PPP investment climate alongside foreign direct investment.

Despite these inflows, governance weaknesses—underinvestment, regulatory uncertainty, weak inter-agency coordination, limited institutional capacity, and unresolved social and environmental issues—have produced mixed outcomes, including delays, cost overruns, and implementation disputes [11]. These issues manifest across both hydropower and oil and gas projects and underscore the need for sector-specific, practice-grounded guidance on how governance mechanisms drive performance in energy PPPs.

International research highlights transparency, accountability, risk management, and stakeholder engagement as core governance themes [12, 13]. However, sector-specific evidence from developing contexts remains limited, and Myanmar's energy PPPs remain underexamined, particularly in studies that compare successful and unsuccessful projects to drive actionable lessons [13, 14]. This gap leaves decision makers uncertain about which governance practices matter most, under what conditions, and how they should be adapted to resource-constrained settings.

This study addresses that gap through a qualitative, comparative multiple-case analysis of four union-level energy PPP projects in Myanmar—two hydropower and two oil and gas projects—purposively selected to include two successful and two unsuccessful cases. Using semi-structured interviews and document review with cross-case synthesis, we assess legal–regulatory arrangements, institutional capacity and coordination, risk allocation and financial structuring, stakeholder engagement, and policy alignment, and identify governance practices linked to stronger performance as well as constraints that undermine delivery.

The contribution is twofold. First, we provide case-verified, sector-specific evidence from a developing-economy context in which long-term energy assets and multi-agency coordination make governance especially consequential. Second, we develop a practice-tested and transferable governance framework that links concrete mechanisms to energy-PPP performance, offering guidance for policy design, contract management, and capacity building in resource-constrained settings [15].

Section 2 synthesizes the literature and isolates the gaps; Section 3 operationalizes them in design, data, and analysis; Section 4 integrates findings with theory and derives implications; Section 5 concludes.

2 Literature Review

This section reviews existing studies on governance in energy infrastructure PPP projects, focusing on Myanmar's energy sector, theoretical frameworks, prior studies, research gaps, and PPP governance. The review is structured into key thematics areas to provide a comprehensive understanding of governance dynamics in PPPs.

2.1 Challenges and Opportunities in Myanmar's Energy Sector

Myanmar's energy sector has historically relied heavily on hydropower, which accounted for nearly half of the country's total installed capacity between 2018 and 2020. However, diversification efforts have led to a gradual increase in natural gas and solar energy contributions [16]. By 2020, solar energy constituted 10.1% of the energy mix, reflecting a shift towards renewable sources. Despite these developments, persistent challenges, including underinvestment, outdated infrastructure, and weak governance, continue to hinder efficient energy production and distribution in the country [17].

Electricity demand has risen steadily, driven by economic growth and urbanization. Over the past decade, annual electricity consumption has increased by 8%, yet approximately 50% of the population lacks reliable access to electricity [18]. Political instability and regulatory inefficiencies exacerbate the demand-supply gap, which widened from 200 MW in 2018 to 600 MW in 2022 [19]. Addressing these issues through PPPs offers a potential pathway to improving infrastructure, provided that robust governance mechanisms are in place to manage risk allocation, stakeholder engagement, and regulatory compliance [20].

2.2 Theoretical and Empirical Perspectives on Public–Private Partnership Governance

Governance in PPPs is grounded in various theoretical frameworks that address the complexities of collaboration between public and private entities. Principal-Agent Theory explores the delegation of tasks from the public sector (principal) to the private sector (agent). This theory emphasizes the challenges posed by information asymmetry, where effective governance structures, such as monitoring and incentive mechanisms, are necessary to align objectives and mitigate risks like moral hazard [21].

Transaction Cost Economics (TCE) focuses on minimizing the costs associated with negotiating, enforcing, and monitoring contracts [22]. In energy infrastructure PPPs, these costs often arise from uncertainties related to risk-sharing, regulatory changes, and operational complexities. Governance mechanisms that ensure contract clarity and efficient dispute resolution are key to enhancing project sustainability.

Institutional Theory (IT) examines how formal rules and informal norms influence organizational behavior, emphasizing that weak institutional frameworks, such as limited regulatory enforcement, pose significant barriers to PPP implementation. Addressing these challenges requires governance structures that promote consistency and predictability in public-private interactions [23].

Governance Theory highlights accountability, transparency, and inclusiveness as foundational principles in decision-making processes [24]. Stakeholder engagement, adaptive leadership, and networked governance are particularly relevant in Myanmar's energy sector, where socio-political and environmental factors intersect. Collaborative governance models based on these principles are essential for building trust, mitigating risks, and ensuring long-term PPP success.

Empirical studies complement these theoretical perspectives by identifying critical governance components that influence PPP outcomes. Transparency, accountability, and stakeholder engagement are often cited as essential for project success [25]. Equitable risk allocation is particularly important in the energy sector, where uncertainties such as demand fluctuations and regulatory changes add complexity [26]. Political support, legal frameworks, and effective communication have also been identified as critical factors for addressing governance challenges in developing economies.

The governance of PPPs in the energy sector poses unique challenges due to high financial stakes, environmental sensitivities, and long project timelines. Shaikh and Randhawa [27] underscore the need for sector-specific governance mechanisms that address stakeholder conflicts and regulatory risks. Xiao [28] identifies transparent risk-sharing and robust stakeholder engagement as key to project success, noting that unresolved governance issues lead to delays and cost overruns. In Southeast Asia, studies highlight the role of governance in balancing development goals with environmental sustainability, particularly in hydropower projects that face community opposition [29].

2.3 Key Components of Effective Public–Private Partnership Governance

PPP governance seeks to align public and private sector interests while safeguarding public welfare and ensuring fair returns for private investors. Effective governance requires well-structured contractual arrangements, comprehensive stakeholder engagement, and transparent communication mechanisms [26]. However, balancing public and private interests within complex regulatory environments remains challenging [25].

Legal compliance plays a pivotal role in reducing risks and fostering investor confidence [30]. In Myanmar, recent reforms have aimed to improve the regulatory environment for PPPs, although enforcement remains inconsistent [31]. Institutional frameworks are crucial for defining roles, streamlining decision-making, and enhancing coordination, particularly in countries like Myanmar with limited PPP experience [25].

Strategic guidelines ensure alignment with project objectives, while innovative financial structuring enhances project viability by improving access to capital markets [32]. Stakeholder engagement is equally important, fostering trust and ensuring that projects meet the needs of all parties involved [33]. Risk management, including the proactive identification and mitigation of risks, is particularly critical in Myanmar's volatile political and regulatory environment [34]. Transparency and accountability further reinforce public trust through clear reporting mechanisms and independent audits [35]. Flexibility and agility are also important, allowing governance structures to adapt to unforeseen challenges while maintaining project objectives [25].

2.4 Regional and Myanmar-Specific Studies

Research on governance in Myanmar's PPPs remains limited, with most studies focusing on broader regional contexts. Southeast Asian PPPs face common challenges, including political transitions and weak institutional frameworks, which are particularly pronounced in Myanmar [36]. PricewaterhouseCoopers (PwC) highlights governance deficiencies in Myanmar's energy sector, including inadequate regulatory frameworks, limited institutional capacity, and weak stakeholder engagement [37]. Political instability exacerbates these issues, increasing the likelihood of project delays and cost overruns. Addressing these barriers requires tailored governance approaches that align with Myanmar's unique socio-political context.

2.5 Comparative Insights from International Literature

Research on PPP governance in transitional economies has highlighted important lessons relevant to Myanmar's context. For instance, studies in Greece emphasize how socio-economic determinants and public attitudes influence the success of renewable energy investments [38, 39]. Onabowale and Mujtaba [40] further demonstrates the effectiveness of financial planning tools in supporting green energy projects, while Rowan et al. [41] shows how citizen engagement shapes electricity usage and trust in renewable systems. Although these studies focus on different contexts, they offer valuable parallels in governance, financial innovation, and participatory strategies for infrastructure development in resource-constrained settings.

2.6 Identification of Research Gaps

Despite the growing body of literature on PPP governance, significant gaps remain. First, empirical studies on governance in Myanmar's energy sector are scarce, limiting the understanding of unique challenges the country faces. Second, comparative analyses of successful and unsuccessful PPP projects are lacking, hindering the ability to derive actionable lessons. Third, Myanmar's political transitions and institutional reforms introduce governance complexities that are not adequately addressed in existing studies. Filling these gaps is critical for developing governance frameworks tailored to Myanmar's energy sector.

To address these gaps, this study employs a qualitative multiple-case design of four union-level energy PPPs in Myanmar—two hydropower and two oil and gas projects, purposively including two successes and two failures—to (i) identify governance practices that most strongly shape performance, (ii) explain the institutional conditions under which they operate, and (iii) synthesize a practice-tested, transferable governance framework for resource-constrained settings.

Research questions and analytical framework, guided by Section 2.6, we ask (R1) which governance practices most strongly shape energy PPP performance; (R2) how and why their implementation and effects differ between successful and unsuccessful cases; and (R3) how the evidence can be synthesized into a transferable governance framework. The analysis uses an input–mechanisms–outcomes framework linking governance domains (inputs) to implementation/ coordination and risk-allocation pathways (mechanisms) and to case performance (outcomes). The next section sets out objectives and methods to operationalize this agenda.

3 Research Methodology

3.1 Research Design

Guided by the gaps in Section 2.6, we use a qualitative multiple-case design to examine governance factors shaping energy-PPP performance at the union-government level. Four PPPs were purposively selected—two hydropower and two oil and gas projects—representing both successful and unsuccessful outcomes to capture institutional variation across the sub-sectors. Interviewee selection followed purposive, maximum-variation criteria, targeting union-level officials directly responsible for the focal projects (procurement, contract management, or regulatory oversight). Eligibility required (i) a direct role in at least one focal case, (ii) at least 10 years' sector experience, and (iii) coverage across hydropower/oil and gas and success/failure cases to ensure case–informant alignment. To reduce institutional influence, we balanced cases and informants across ministries and success/failure outcomes and assured confidentiality to decouple responses from organizational loyalties. In operational terms, inputs are mapped to the 82 codebook variables; mechanisms are traced through coded implementation quality, interagency coordination, and risk-allocation pathways; outcomes are assessed using qualitative impact levels (see Section 3.5.1) and cross-case influence levels (see Section 3.5.2).

Data collection combined semi-structured interviews and document review. Four senior officials from union-level line ministries directly involved in the projects participated in 60–90-minute interviews. The unit of analysis is union-level governance mechanisms, and interviewees were selected as elite informants because they held direct operational responsibility for procurement, contract management, or regulatory oversight in the focal PPP projects rather than providing general opinions. The interview guide, derived from a systematic literature review and pilot-tested for contextual fit, addressed legal frameworks, institutional arrangements, strategic guidelines, and financial and risk-management processes.

Analysis followed in-case and cross-case procedures using established protocols. Sample adequacy is justified by the study's narrow aim, elite informants, and strong triangulation with contracts, procurement files, and policy documents; no new first-order codes emerged after the third interview, indicating thematic sufficiency for the targeted union-level phenomena. To make the assessment of data sufficiency transparent, interview saturation was evaluated during the coding process. In this study, a new first-order code refers to a governance practice or mechanism that had not previously appeared in the predefined codebook derived from the literature review. After each interview, transcripts were coded and compared with earlier interviews to identify whether additional first-order codes emerged. When subsequent interviews yielded only confirmations or elaborations of existing codes without introducing new ones, thematic sufficiency was considered achieved for the targeted union-level governance mechanisms.

A brief coding summary illustrates this pattern. Interview 1 generated 18 first-order codes related to governance practices. Interview 2 added 6 additional codes. Interview 3 produced only 2 minor refinements without introducing substantially new governance mechanisms. Interview 4 confirmed previously identified themes and provided additional contextual evidence rather than new codes. This pattern indicated that the key governance mechanisms relevant to the study scope had already been captured. Triangulation of interview and documentary evidence enhanced contextual depth, credibility, and reliability [42, 43].

3.2 Systematic Literature Review Process

A systematic literature review following Leidecker and Bulman guided the variable set [44]. From an initial corpus, we retained 50 studies using inclusion criteria requiring that a governance construct be (i) sector-relevant to energy PPPs, (ii) operationalizable at the union/project level, and (iii) recurrent across three or more sources or flagged by experts as critical. Duplicate or overlapping constructs were merged using a codebook (name, definition, unit/scale, source), focusing on Critical Success Factors, risk management, and governance best practices [45]. The outcome was four components, sixteen sub-components, and eighty-two variables aligned to energy-PPP governance practice.

Each variable in the codebook includes a plain-language definition and observable evidence (document/interview anchors) to support reproducibility.

To integrate institutional perspectives, the literature review also examined frameworks that analyze the interdependence between ministries and regulatory bodies. Drawing from Baniya and Giurco [9], who used similar review strategies to identify how institutional structures shape climate action implementation in Nepal, this review informed the analytical lens for understanding institutional arrangements within Myanmar's PPP system. No numeric weighting was applied during variable construction; selection relied on these criteria and expert consensus.

3.3 Expert Validation

Two iterative rounds with national PPP experts—officials and analysts from energy-infrastructure agencies and the Central Statistical Organization—refined the framework. Experts reviewed a codebook (variable name, definition, evidence source) for face and content validity and resolved boundary issues. Acceptance required consensus on definition, measurability, and the document/interview anchor; non-consensus items were merged or dropped. The process yielded four components, sixteen sub-components, and 82 variables with stable operational definitions, as summarized in Table 1. Experts also underscored institutional capacity gaps and cross-agency asymmetries, echoing multi-level governance concerns in comparable contexts.

Table 1 presents the proposed governance framework for PPP projects in Myanmar, including the four governance components, their associated sub-components, and the corresponding operational variables.

Table 1. Proposed governance framework for Public–Private Partnership (PPP) project in Myanmar

Governance Component	Governance Sub–Component	Variables	Governance Component	Governance Sub–Component	Variables
I. Legal and Regulatory	1.0-PPP Law and Regulations	1.1 PPP legal communication	II. Institutional framework	7.0 Line ministries' contribution to PPP project	7.1 Ministry participation
		1.2 PPP legal compliance			7.2 Ministry data provision
		1.3 Regulatory collaboration			7.3 Ministry's PPP integration
		1.4 Regulatory agility			7.4 Ministry regulatory compliance
	2.0 Procurement law compliance	2.1 Procurement process alignment		8.0 Regulatory agencies compliance monitoring	8.1 Regulatory agency audits & inspections
		2.2 Procurement record compliance			8.2 Regulatory agency proactiveness
		2.3 Equal and fair opportunity			8.3 Regulatory agency collaboration
		2.4 Procurement policy alignment			8.4 Regulatory agency accountability
	3.0 Contract adherence	3.1 Contract compliance		9.0 Local government engagement	9.1 Local government contribution
		3.2 Contractual clarity			9.2 Local government policy alignment
		3.3 Contract audits			9.3 Local government engagement stakeholders
		3.4 Dispute resolution			9.4 Local government project selection

Governance Component	Governance SubComponent	Variables	Governance Component	Governance SubComponent	Variables
II. Institutional framework	4.0 Intellectual property rights	4.1 IPR risk assessment	III. Strategies and guidelines	10.0 PPP unit skills and capacity	10.1 PPP unit technical expertise
		4.2 IPR protection measure			10.2 PPP unit staff training
		4.3 IPR enforcement collaboration			10.3 External expert collaboration
		4.4 IPR research and development focus			10.4 PPP unit resources and tools
	5.0 Regulatory oversight	5.1 Information disclosure		11.0 National PPP policy adherence	11.1 National PPP policy integration
		5.2 Compliance personnel assessment			11.2 National PPP policy utilization
		5.3 Conflict of interest prevention			11.3 National PPP policy for transparency
		5.4 Compliance training			11.4 National PPP policy risk allocation
	6.0 PPP unit functionality	6.1 PPP unit coordination			11.5 National PPP policy competition enhancement
		6.2 PPP unit stakeholder engagement			11.6 National PPP policy alignment
		6.3 PPP unit organizational structure			11.7 National PPP policy training
		6.4 PPP unit compliance assurance			
III. Strategies and guidelines	12.0 PPP preparation guidelines	12.1 Guidelines standardization	IV. Financial arrangement and risk management	14.0 Financing source exploration in PPP project	14.1 Alternative financing source
		12.2 Project appraisal enhancement			14.2 Incentives for investment in PPP
		12.3 Guideline-based PSC development			14.3 Innovative financial instruments in PPP
		12.4 Guidelines utilization in impact assessment			14.4 Financial collaborations with international institutions
		12.5 Guidelines utilization document preparation			14.5 Financial partnerships for long-term financing
		12.6 Guidelines incorporation			14.6 Financial structuring for private investments

Governance Component	Governance SubComponent	Variables	Governance Component	Governance SubComponent	Variables
		12.7 Guidelines for technical feasibility			14.7 Sector-specific financing models for sustainable funding
		13.1 Contract management guidelines			15.1 Taxation incentives for private sector investment
	13.0 PPP execution guidelines	13.2 Procurement guidelines compliance		15.0 Taxation and fiscal framework for private sector participation	15.2 Taxation framework for transparency and predictability
		13.3 Communication & dispute resolution guidelines			15.3 Risk-sharing mechanisms in taxation and fiscal framework
		13.4 Performance monitoring & evaluation guidelines			15.4 Taxation and fiscal framework for innovative financing
		13.5 Financial transparency and accountability guidelines			15.5 Taxation and fiscal framework design to address barriers for private sector participation
		13.6 Asset and responsibility transfer guidelines			15.6 Alignment of taxation and fiscal framework with international best practices
		13.7 Guidelines for effectiveness in critical project elements			15.7 Tax incentives evaluation
IV. Financial arrangement and risk management	16.0 Risk allocation mechanisms for fair distribution of risks in PPP	16.1 Clarity and documentation of risk allocation mechanisms			
		16.2 Implementation and enforcement of risk allocation mechanisms			
		16.3 Design and mitigation of risk allocation mechanisms			
		16.4 Alignment of risk allocation mechanisms			

Governance Component	Governance SubComponent	Variables	Governance Component	Governance SubComponent	Variables
		16.5 Review and update of risk allocation mechanisms 16.6 Effectiveness of risk allocation mechanisms 16.7 Communication and understanding of risk allocation mechanisms			

Note: IPR = Intellectual Property Rights.

3.4 Application of Interviews

Semi-structured interviews elicited qualitative evidence on governance mechanisms, institutional behavior, and implementation practices, focusing on both formal structures and informal dynamics (e.g., bureaucratic rivalries, political incentives, inter-agency negotiation). Reliability and validity were supported by a protocolized, pilot-tested guide; standardized interviewing; audio-recording with verbatim notes; codebook-guided coding with stability checks; peer debriefs; and a chain-of-evidence (case files, code memos, document logs).

Data were triangulated with contracts, guidelines, and legal/policy documents, and factual statements were cross verified against documentary sources before inclusion. Negative-case searches during cross-case synthesis challenged emerging propositions. Interview materials were then coded for in-case analysis, with patterns informing the cross-case evaluation. To mitigate respondent bias and institutional influence, we assured anonymity, balanced respondents across agencies and success–failure cases, and triangulated sensitive claims across institutions.

The data-collection period spanned 5 February–22 March, 2024, with three interview rounds per case and subsequent document receipt. Table 2 presents details of each case, including interviewee roles and interview periods.

Table 2. Case study of Public–Private Partnership (PPP) projects

Code	Type	Case Nature	Performance	Start	Interviewee Position	Interviewee Service	Interview Meeting 1st	Interview Meeting 2nd	Interview Meeting 3rd	Received Document
HP1	Hydropower	Green field	Success	2002	Deputy director	10 years above	5 February 2024	13 February 2024	28 February 2024	14 March 2024
HP2	Hydropower	Green field	Unsuccess	2010	Assistant director	10 years above	5 February 2024	14 February 2024	2 March 2024	20 March 2024
OG1	Oil-and gas (Pipeline)	Green field	Success	2010	Assistant director	10 years above	6 February 2024	15 February 2024	5 March 2024	18 March 2024
OG2	Oil-and gas (Drilling)	Greenfield	Unsuccess	2009	Assistant director	10 years above	6 February 2024	16 February 2024	8 March 2024	22 March 2024

3.5 Case Study Analysis

To clarify the analytical procedure, the study followed a three-step workflow. First, the 82 governance variables defined in the codebook were mapped to case evidence using interview transcripts and project documents. Each variable was assessed for its implementation status within each case based on convergent evidence from interviews, contracts, procurement records, and policy documents. Second, qualitative impact labels (high positive, medium positive, no impact, medium negative, and high negative) were assigned using the structured rubric described in Section 3.5.1, relying on triangulated evidence rather than numerical scoring. Third, cross-case comparison of the four projects was conducted to determine influence levels (critical, high, moderate, and low) by identifying patterns of presence in successful cases and absence in unsuccessful cases.

For example, the variable PPP Legal Compliance (1.2) was examined using interview evidence from union-level officials and documentary evidence such as contracts and regulatory guidelines. In successful cases (HP1 and OG1), strong compliance mechanisms were observed, while unsuccessful cases showed fragmented enforcement. Based on this evidence pattern, the variable received a high positive impact label in successful projects and a high negative impact label in unsuccessful projects, leading to its classification as a Critical influence factor in the cross-case analysis.

3.5.1 Application of in-case qualitative impact analysis

This analysis examined how specific governance variables shaped outcomes in individual PPP projects. Stake's case study methodology guided the interpretation of how governance practices contributed to success or failure [42].

We applied a qualitative rubric to classify the effects of each variable: High Positive denotes critical, outcome-advancing effects corroborated by multiple sources; Medium Positive denotes beneficial but non-critical improvements; No Impact denotes a neutral or negligible effect; Medium Negative denotes manageable detriments (e.g., minor delays or cost increases); and High Negative denotes severe adverse effects (e.g., major delays, cost escalation, or quality compromise). Labels are ordinal and non-numeric, based on triangulated interview and document evidence. The rubric is interpretive and evidence-based, intended to support structured comparison across cases rather than numerical measurement of effects.

This structured scheme aligns with Yin's principles for contextual case study research [43]. Critical factors such as legal compliance and risk management were strongly associated with project success, echoing findings by Pinto and Slevin [16]. Causal identification followed a process-tracing approach: for each variable, we coded (i) implementation status, (ii) the immediate mechanism it activated (e.g., monitoring intensity, timing of renegotiation, or data sharing), and (iii) the observed performance effect (delays, costs, quality deviations, and disputes), requiring at least two independent, convergent sources per claim.

The results identified practices that either advanced or impeded performance, consistent with evidence on large-scale infrastructure PPPs reported by Maqbool and Schmidt [46, 47]. The framework clarifies the role of governance in shaping effectiveness at the project level.

Assessment of impact level. The PPP Legal Communication (1.1) variable had a high positive impact in HP1 (success) because a formal process communicated legal requirements to stakeholders. Clear, timely notices reduced delays, promoted transparency, and directly advanced strategic objectives.

The Implementation and Enforcement of Risk Allocation Mechanisms (16.2) showed a medium positive impact in OG1 (successful) owing to adaptive risk monitoring through regular progress reviews. Although formal procedures were absent, proactive coordination partially mitigated risks and improved operational efficiency, underscoring the value of flexibility.

The National PPP Policy Integration (11.1) had no impact on OG1 (successful) because the project predated the policy's formulation; nevertheless, established governance mechanisms sustained performance.

The Alignment of Risk Allocation Mechanisms (16.4) had a medium negative impact in HP2 (unsuccessful) due to the absence of formal policies for fair risk allocation. Planning-stage discussions proved insufficient, culminating in disputes and inefficiencies.

The PPP Unit Coordination (6.1) had a high negative impact in OG2 (unsuccessful) because no formal coordination system existed. Communication gaps among the PPP Unit, the JV company, and stakeholders delayed discussions and approvals, highlighting the need for structured frameworks.

Table 3. Results of implementation and impact level of governance variables

Variables	Successful Cases				Unsuccessful Cases			
	HP1		OG1		HP2		OG2	
	Implementation	Impact Level	Implementation	Impact Level	Implementation	Impact Level	Implementation	Impact Level
1.1 PPP legal communication	Yes	High positive	Yes	High positive	Partially	Medium negative	Yes	Medium positive
1.2 PPP legal compliance	Partially	Medium positive	Yes	High positive	No	High negative	Yes	High positive
1.3 Regulatory collaboration	Yes	High positive	Partially	Medium positive	Yes	Medium positive	No	High negative
1.4 Regulatory agility	Yes	High positive	Yes	High positive	Yes	Medium positive	Yes	Medium positive

Notes: Definitions & Abbreviations: Implementation codes: Yes = fully implemented; Partially = implemented with gaps; No = not implemented. Impact scale and polarity: High/Medium/Low Positive (strong/moderate/mild beneficial effect), Neutral (no material effect), High/Medium/Low Negative (adverse effect). Case codes: HP = Hydropower; OG = Oil and Gas.

Table 3 summarizes the implementation status and impact levels for variables in the PPP Law and Regulation sub-component (1.0) of the Legal and Regulatory Governance component. Overall, the analysis spans 82 variables across 16 sub-components within four governance components, indicating each variable's contribution to PPP outcomes in practice.

3.5.2 Application of cross-case qualitative influence level analysis

This cross-case qualitative influence level analysis evaluated governance variables across multiple PPP projects to surface commonalities and differences in governance practice. Variables were grouped into four influence levels: Critical, High, Moderate, and Low. Critical factors tended to drove success when implemented; High factors were important but allowed some flexibility. Moderate factors added value without being essential, while Low factors showed context-dependent effects [48]. Influence levels (Critical/High/Moderate/Low) are ordinal, qualitative syntheses of convergent case patterns; no numerical scaling or statistical weighting is used. Cross-case generalization used a necessary-and-sufficient rule: A practice is classified as Critical when its absence co-occurs with high negative effects in both unsuccessful cases and its presence co-occurs with positive effects in both successful cases.

The approach follows the methodology of Suhana et al. [45], which emphasizes comparative case studies for broader inference, and draws on Landis's work on governance in large-scale infrastructure [49]. Comparing successful and unsuccessful cases enabled the identification of practices that are universally important and those contingent on project conditions.

Assessment of influence level. Governance variables were assigned to Critical, High, Moderate, or Low based on observed effects on PPP outcomes. Each variable was judged for its contribution to success or propensity to create challenges.

The PPP Legal Compliance (1.2) was a Critical factor. In successful projects, clear compliance mitigated risk, increased transparency, and aligned actions with governance frameworks, yielding high positive effects. Its absence in unsuccessful projects produced high negative effects, underscoring its indispensability for strategic objectives and failure prevention.

The Contract Compliance (3.1) showed High influence. In successful projects, adherence to terms reduced disputes and delays, producing high positive effects. In unsuccessful cases, weaker adherence produced only medium positive effects, indicating its role in clarity, accountability, and operational efficiency.

The Local Government Project Selection (9.4), a Moderate factor, aligned objectives with strategic priorities, generating high effects in the successful case. In unsuccessful cases, its absence caused medium to high negative effects, indicating that effective selection supports strategic fit and execution.

Table 4 categorizes governance variables by influence level across projects. The analysis spans 82 variables across 16 sub-components under four governance components, clarifying each variable's role in shaping PPP outcomes and informing prioritization for governance reforms and oversight.

Table 4. Results of qualitative influence level analysis

Governance Component	Sub–Component	Variable	Cross-Case Analysis	Comment	Influence Level
I. Legal and regulatory framework	4.0 Intellectual property rights	4.4 IPR research and development focus	Common: Not implemented in any cases; Differences: Medium negative impact in successful cases; none in unsuccessful cases.	Absence of an R&D focus limits innovation and competitiveness in successful cases; it has no impact in projects relying on external technologies.	Low

Governance Component	SubComponent	Variable	Cross-Case Analysis	Comment	Influence Level
II. Institutional framework	6.0 PPP unit functionality	6.2 PPP unit stakeholder engagement	Common: Present in successful cases; partial or absent in unsuccessful cases; Differences: High positive in successful cases; medium negative to high negative in unsuccessful cases.	Stakeholder engagement fosters trust and cooperation. Partial or absent engagement in unsuccessful cases correlates with poor collaboration and conflicts.	High
III. PPP strategies and guidelines	12.0 PPP preparation guidelines	12.3 Guideline-based public sector comparator (PSC) development	Common: Partial or absent in most cases; Differences: Medium positive in successful cases; medium negative to high negative in unsuccessful cases.	PSC development supports value-for-money assessments. Absence significantly impacts cost efficiency and stakeholder confidence in unsuccessful cases.	Moderate
IV. PPP financial arrangement and risk management	14.0 Financing source exploration in PPP project	14.3 Innovative financial instruments in PPP	Common: Absent in all cases; Differences: Medium negative in successful cases; high negative in unsuccessful cases.	Innovation in financial instruments fosters cost efficiency and risk management. Absence introduces inefficiencies and weakens project resilience.	Critical

Notes: Definitions & Abbreviations: PPP = Public-Private Partnership; IPR = Intellectual Property Rights; PSC = Public Sector Comparator. Influence level (ordered): Critical > High > Moderate > Low. Cross-case analysis uses "Common" for patterns shared across cases and "Differences" for directional contrasts between successful and unsuccessful cases.

3.6 Key Factors

The study identified 22 critical factors essential for the successful governance of energy infrastructure PPP projects in Myanmar. These factors were strongly associated with project success, emphasizing their importance in achieving project goals. Their absence in unsuccessful projects indicated the risks of inadequate governance. Table 5 outlines these factors, detailing their roles and relevance to PPP success in Myanmar's energy sector.

Table 5. Key factors of Public–Private Partnership (PPP) governance Myanmar

No.	Governance Component	Sub-Governance Component	Variable	Influence Factor	Explanation for Inclusion as Key Factor	Evidence Source
1	I. Legal and regulatory	1.0-PPP law and regulations	1.2 PPP legal compliance	Critical	Prevents failure; aligns with governance; cuts risk.	I+D
2		2.0 Procurement law compliance	2.3 Equal and fair opportunity	Critical	Avoids disputes, builds trust.	I+D
3		3.0 Contract adherence	3.3 Contract audits	Critical	Boosts accountability/risk control; gaps cause inefficiency	I+D
4		4.0 Intellectual property rights	4.2 IPR protection measure	Critical	Preserves investor confidence in tech-heavy PPP.	D
5		5.0 Regulatory oversight	5.2 Compliance personnel assessment	Critical	Ensures consistent execution; absence breeds errors	I
6	II. Institutional framework	6.0 PPP unit functionality	6.1 PPP unit coordination	Critical	Speeds delivery; weak links cause delays mismanagement	I
7		7.0 Line ministries' contribution to PPP project	7.2 Ministry data provision	Critical	Informs decisions; lack = delays, opacity.	I+D
8		8.0 Regulatory agencies compliance monitoring	8.2 Regulatory agency proactiveness	Critical	Mitigates emerging risks; absence hurts outcomes.	I
9		9.0 Local government engagement	9.2 Local government policy alignment	Critical	Strategic coherence; Misalignment → conflict inefficiency	I
10		10.0 PPP unit skills and capacity	10.4 PPP unit resources and tools	Critical	Enables operations; shortages raise risk/inefficiency.	I
11	III. Strategies and guidelines	11.0 National PPP policy adherence	11.3 National PPP policy for transparency	Critical	Builds trust/accountability; opacity invites mismanagement	D
12		12.0 PPP preparation guidelines	12.5 Guidelines utilization document preparation	Critical	Traceability & alignment; poor records cut accountability.	D
13		13.0 PPP Execution guidelines	13.1 Contract management guidelines	Critical	Drives day-to-day performance; weakness derails results.	I+D
14			13.3 Communication & Dispute resolution guidelines	Critical	Maintains alignment; absence disrupts.	I
15			13.5 Financial transparency and accountability guidelines	Critical	Safeguards governance; weak oversight risks failure.	D

No.	Governance Component	Sub-governance Component	Variable	Influence Factor	Explanation for Inclusion as Key Factor	Evidence Source
16	IV. Financial arrangement and risk management	14.0 Financing source exploration in PPP project	14.3 Innovative financial instruments in PPP	Critical	Financing stability: absence hurts sustainability.	D
17			14.5 Financial partnerships for long-term financing	Critical	Attracts private capital; lack deters funding.	D
18		15.0 Taxation and fiscal framework for private sector participation	15.1 Taxation incentives for private sector investment	Critical	Expands viable funding options; absence constrains	D
19			15.4 Taxation and fiscal framework support for innovative financing	Critical	Consistent governance: ambiguity hinders execution.	D
20		16.0 Risk allocation mechanisms for fair distribution of risks in PPP	16.1 Clarity and documentation of fisk allocation mechanisms	Critical	Consistent governance: Ambiguity hinders execution.	I+D
21			16.5 Review and update of risk allocation mechanisms	Critical	Keeps practices current; neglect compounds inefficiency.	I
22			16.7 Communication and understanding of risk allocation mechanisms	Critical	Shared understanding: poor comms undermine governance.	I

Notes: Evidence sources supporting each critical factor are indicated as follows: I = interview evidence, D = documentary evidence, and I+D = triangulated evidence from both interviews and documentary sources.

These findings expand the scope of PPP governance research, addressing overlooked areas such as tailored legal frameworks, institutional capacity, intellectual property protections, financial mechanisms, and stakeholder collaboration. They provide actionable insights for developing governance frameworks tailored to resource-constrained, context-specific challenges such as those in Myanmar. These cross-case results motivate Section 4, which links the 22 factors to mechanisms (Principal–Agent (PA), TCE, and Institutional/Network) and derives implications.

4 Discussion

Building directly on Section 3's findings, this section links variables to mechanisms and outcomes and integrates the results with theory. For theoretical integration, drawing on PA, TCE, IT, and Network/Collaborative governance perspectives, the 22 factors operate via three mechanisms: (1) reducing information asymmetry and aligning incentives (PA), e.g., legal compliance (1.2), transparent procurement (2.3), and contract audits (3.3); (2) lowering ex ante/ex post transaction and coordination costs (TCE), e.g., Communication & Dispute Resolution guidelines (13.3), clarity, period, review, and communication of risk allocation (16.1, 16.5, 16.7); and (3) strengthening formal rules and inter-organizational capacity (IT/Network), e.g., PPP Unit Coordination (6.1), Regulatory Agency Proactiveness (8.2), and national policy transparency (11.3). This anchoring clarifies why each factor affects performance in capital-intensive, long-horizon energy PPPs.

The causal propositions are as follows. Proposition 1 (P1), grounded in PA: Legal compliance, transparent procurement, and contract audits reduce information asymmetry and moral hazard, thereby lowering disputes and renegotiations and shortening delivery timelines (the evidence suggests this pattern in HP1 and OG1, while it was not evident in HP2 and OG2). P2 (TCE): Communication and dispute-resolution routines, together with clear and periodically reviewed risk-allocation rules, reduce ex ante and ex post bargaining and hold-up costs, leading to fewer stoppages and smoother approvals; the cases indicate this influence in HP1 and OG1, while it was weaker in HP2 and absent in OG2. P3 (Institutional/Network): PPP-unit coordination, regulatory proactiveness, and policy transparency build interagency capacity and predictable routines, resulting in faster decisions and more coherent implementation; the pattern indicates stronger effects in HP1 and OG1 and weaker influence in HP2 and OG2.

4.1 Key Factors in Public–Private Partnership Governance of Energy Infrastructure

Recent studies from Greece and transitional economies highlight how socio-economic conditions and public perception shape renewable energy investment and PPP frameworks [39, 50]. Their insights, though contextually distinct, parallel Myanmar's evolving energy sector.

This study identifies 22 governance factors influencing energy infrastructure PPP performance in Myanmar, summarized in Table 5. These factors combine established governance principles with context-specific dynamics operating within the union-level institutional structure. Because all four cases fall under the Union Government—constitutionally responsible for national-scale energy infrastructure since 2008—the analysis adopts a vertical federal governance lens, with limited regional government involvement.

4.1.1 Linking governance to energy infrastructure Public–Private Partnerships

The selected PPP projects—two hydropower and two oil and gas projects—are capital-intensive, long-term ventures that depend on coordinated federal institutions. Governance failures can disrupt the lifecycle, from planning and procurement to construction and operation. The governance variables capture cross-sector concerns and energy bottlenecks, clarifying the PPP–energy infrastructure nexus.

Several factors align with international literature and are salient in the energy sector because of their scale, risk, and technical complexity. Equal and Fair Opportunity in Procurement (2.3) and Compliance Personnel Assessment (5.2) ensure transparency and curb elite capture in awarding concessions [51, 52] (PA: reduces information asymmetry; TCE: lowers monitoring/enforcement costs). Similarly, PPP Unit Coordination (6.1) and Regulatory Agency Proactiveness (8.2), supporting Garg [53], reduce delays in power generation and notably pipeline development, suggesting institutional capacity as an important contributor to project performance (IT/Network: strengthens interagency capability and reduces coordination failure).

Moreover, Ministry Data Provision (7.2) appears important for planning and monitoring technically demanding energy assets, with reliable information serving as a governance foundation (PA: mitigates hidden information; TCE: cuts verification costs). Failures in projects lacking timely, accurate data suggest the sector's sensitivity to information asymmetries, which distort risk assessment and misalign project scopes.

4.1.2 Role of federal institutions in driving energy Public–Private Partnerships

Because energy PPPs in Myanmar are centrally run, the Union Government—through line ministries and sectoral agencies—is the primary actor for policy design, procurement, and monitoring. Local governments play limited roles; however, Local Government Policy Alignment (9.2) supports socio-political coherence and public acceptance, especially for land access and resettlement in hydropower (IT/Network: vertical coherence for land/social issues). This clarifies vertical governance: strong central control with limited, strategic local support.

Furthermore, weak institutional synergy among federal entities undermines performance. Failures in Communication and Dispute Resolution Guidelines (13.3) show how inter-agency conflict and bureaucratic silos delay decisions, timely approvals, and implementation, a concern widely noted for large-scale infrastructure PPPs [54] (TCE: lowers renegotiation/hold-up costs).

4.1.3 Myanmar-specific governance gaps

Contrary to global PPP models, this study identifies governance elements salient in Myanmar's federal context yet underexplored in broader literature. PPP Legal Compliance (1.2) and Contract Audits (3.3) are critical given the evolving legal landscape and weak enforcement. While Pinilla-De La Cruz et al. [55] stresses robust legal foundations, this study suggests that absent energy-aligned mechanisms create contractual ambiguities and delay implementation.

The absence of effective Intellectual Property Rights (IPR) Protection Measures (4.2) across all four cases reveals an oversight in governing energy technologies and know-how transfers, particularly in hydropower and gas assets with proprietary designs or foreign inputs. This gap, seldom addressed in PPP literature, appears as a confidence-eroding factor for international partners [56].

Likewise, Guidelines Utilization for Document Preparation (12.5) and National PPP Policy for Transparency (11.3) are critical in Myanmar's energy PPPs, where nonstandard procedures and opaque approvals impede governance. While Rybnicek et al. [30] emphasizes guidelines and transparency, systematic integration through documentation and national policy remains insufficient. Successful projects applied these mechanisms to improve coordination and build stakeholder trust, whereas their absence in unsuccessful cases led to fragmentation and higher risks.

4.1.4 Financial and risk management in energy Public–Private Partnerships

The energy sector's large upfront capital needs make financial governance decisive. The presence of Innovative Financial Instruments (14.3) and Taxation and Fiscal Framework Support for Innovative Financing (15.4) in successful projects suggests the value of adaptive tools beyond conventional funding, especially in Myanmar's constrained public finances and shallow capital markets (TCE: minimizes financing and renegotiation cost). These findings complement Bai et al. [57] on financial innovation while extending their relevance to politically transitioning, resource-scarce settings in practice.

Communication and Understanding in Risk Allocation Mechanisms (16.7) appeared important for structuring long-term contracts that span decades and multiple stakeholder layers (PA/TCE: hazard allocation clarity and moral-hazard control). Misunderstandings produced trust deficits, delays, and inefficient renegotiations, which were patterns more pronounced in unsuccessful cases, indicating the importance of integrated, shared risk communication across federal PPP institutions.

Financial governance also enabled sustainable implementation. Financial Partnerships for Long-Term Financing (14.5) and Taxation Incentives for Private Sector Investment (15.1) improved bankability and investor confidence, consistent with Bai et al. [57] (PA: aligns private effort with public goals). Equally important, risk-allocation measures—Clarity and Documentation (16.1) and Review and Update (16.5)—reduced disputes and increased resilience, aligning with Swanson and Sakhrani (2020). Contract Management Guidelines (13.1) supported operational continuity and accountability during execution. Institutional capacity, notably PPP Unit Resources and Tools (10.4), was essential in resource-limited environments. While Garg [53] emphasizes general capacity concerns, this study shows that practical access to tools and systems directly shaped coordination and success.

Overall, the results widen PPP governance research by addressing overlooked areas—tailored legal frameworks, institutional capacity, intellectual property protection, financial mechanisms, and stakeholder collaboration—and offer actionable guidance for building governance frameworks suited to resource-constrained, context-specific challenges such as those in Myanmar.

4.2 Major Challenges in Public–Private Partnership Governance of Energy Infrastructure Project in Myanmar

Cross-case analysis of successful and unsuccessful energy PPPs reveals systemic governance challenges in Myanmar. They are amplified by the Union Government's centralized responsibility for large-scale projects under the constitution. This top-down structure shapes legal compliance, institutional coordination, and policy execution, clarifying the hierarchical dynamics of PPP governance.

Within the Legal and Regulatory Framework, weak PPP Legal Compliance (1.2) and limited Regulatory Collaboration (1.3) appear to have acted as important barriers. OG2 suffered fragmented compliance, whereas HP1 showed that robust procedures stabilized the project. Insufficient Regulatory Agility (1.4) further constrained adaptation to evolving conditions. Persistent neglect of IPR Risk Assessment (4.1) and Research and Development Focus (4.4) curtailed innovation, exposing a mismatch between technological needs and safeguards.

Risk Allocation Mechanisms also showed critical flaws. Missing Clear Documentation (16.1), misaligned frameworks (16.4), and weak Communication and Understanding (16.7) appeared to weaken project outcomes. HP1 achieved partial gains from incomplete measures, but OG2 faced high risks from the absence of structured risk-sharing.

Institutional Capacity gaps were pervasive. HP1 benefited from PPP Unit Technical Expertise (10.1) and Staff Training (10.2); OG2 lacked both, with high negative impacts. Irregular use of External Experts (10.3) further weakened technically demanding projects.

Financial Structuring was acute in resource-constrained settings. Unsuccessful cases lacked Innovative Financial Instruments (14.3) and Alternative Funding Sources (14.1). HP1 leveraged Financial Collaborations with International Institutions (14.4), suggesting that tailored strategies can support sustainability and attract investment. Absent risk-sharing in taxation (15.3) reduced private-sector confidence.

Stakeholder Engagement varied. HP1's strong Stakeholder Engagement Framework (9.3) appeared to improve alignment, while OG2's failure to engage led to disputes and delays. Absent Communication and Dispute Resolution Guidelines (13.3) heightened tensions in unsuccessful cases.

Finally, Policy Alignment with National PPP Policies (11.6) suggested differences in performance. HP1 aligned, contributing to strategic coherence; OG2 did not, creating inconsistencies. Underuse of guidelines during preparation (12.2) and financial structuring (14.6) compounded these issues—even where partial success occurred.

Collectively, these challenges show that legal rigor, capable institutions, fit-for-purpose risk allocation, innovative financing, purposeful stakeholder processes, and clear policy alignment are prerequisites for reliable energy-PPP performance in Myanmar.

4.3 Addressing Important Issues with Key Factors Findings

This study identifies three nuanced governance insights critical to understanding the PPP energy infrastructure nexus in Myanmar:

4.3.1 Insight 1: Centralized governance requires stronger institutional coordination

The Union Government's centralized oversight of energy PPPs ensures consistent national direction, yet weak coordination among ministries and agencies undermines delivery. Deficits in PPP Unit Coordination (6.1), limited Data Provision (7.2), and insufficient Regulatory Proactiveness (8.2) produce fragmented decisions and delays. Successful projects, by contrast, show that robust vertical and horizontal collaboration strengthens accountability, improves communication, and enables smoother execution across agencies and tiers [58].

4.3.2 Insight 2: Policy-project disconnect undermines implementation

A persistent gap separates national PPP policies from project-level practice. Although national guidelines stress transparency and due diligence, critical variables—Guidelines Utilization for Document Preparation (12.5) and National PPP Policy for Transparency (11.3)—were under-implemented in unsuccessful cases, generating inconsistent procedures and stakeholder mistrust. Projects that explicitly aligned policy directives with daily operations achieved greater coherence and stronger performance outcomes.

4.3.3 Insight 3: Financial innovation and risk communication drive Public–Private Partnership viability

In resource-constrained contexts, PPP viability hinges on innovative finance and clear risk-sharing. Innovative Financial Instruments (14.3), Taxation Incentives (15.1), and Communication and Understanding of Risk Allocation Mechanisms (16.7) appear important. Where flexible tools and transparent allocation frameworks were absent, projects struggled to attract investment and sustain long-term performance. Conversely, diversified financing and a shared understanding of responsibilities improved resilience and investor confidence.

Together, these insights indicate that strengthening inter-ministerial coordination, closing the policy–practice gap, and embracing adaptive finance are fundamental to improving energy-PPP governance in Myanmar and comparable developing economies.

International evidence reinforces these findings. Studies emphasize financial innovation, citizen engagement, and regional energy planning during transitions toward renewables. Research from Southern Europe shows how public willingness to invest, fiscal incentives, and decentralized participation shape the viability of green-energy projects [39, 59]. These perspectives offer practical lessons for Myanmar's evolving energy sector as it advances decentralized and renewable initiatives alongside large-scale PPPs. Such lessons remain practically relevant.

4.4 Recommendation for Improving Public–Private Partnership Governance

To improve PPP governance in Myanmar's energy infrastructure, reforms should begin with the legal and regulatory framework. Enacting a dedicated PPP Act defining stakeholder roles and risk-sharing responsibilities would promote consistency and clarity. Digitalizing procurement via an e-platform would widen competition and transparency in tendering.

Risk allocation requires reinforcement through clear documentation, periodic framework reviews, and training to build shared understanding among stakeholders. Effective risk communication reduces misunderstandings and mitigates implementation delays.

Institutional capacity should be strengthened by establishing a central PPP coordination unit with adequate staffing, digital tools, and interministerial communication channels. Capacity-building must emphasize financial modeling, project appraisal, and contract management. Independent audit mechanisms enhance regulatory oversight, while streamlined interministerial data-sharing protocols reduce delays and gaps.

Innovative financing is vital in resource-constrained settings. Myanmar should deploy instruments such as green bonds and blended finance, complemented by fiscal incentives (e.g., tax holidays) to attract private capital. Transparency in financial reporting, regular audits, and public disclosure of financial data would build investor and development-partner confidence.

Stakeholder engagement is a priority. Formal engagement forums and feedback mechanisms align expectations, while embedding dispute-resolution protocols in project design mitigates conflict. Communication must be continuous, inclusive, and structured to sustain trust across the project lifecycle.

Governance practices must align with national PPP policies. Practical integration at the project level—through standardized templates, checklists, and reporting frameworks—reduces fragmentation. Training and implementation support should ensure policies are actively used, not symbolic.

Finally, robust intellectual property rights protection will strengthen investor confidence, particularly for technology-intensive projects. This requires the development of enforceable IPR mechanisms to safeguard innovation and limit knowledge leakage.

By systematically implementing these improvements, Myanmar can enhance the performance, sustainability, and credibility of energy PPPs, offering a replicable model for other developing economies facing comparable institutional constraints.

4.5 Implication for Theory and Practice

The findings offer both theoretical and practical implications for the governance of PPPs in the energy sector in developing countries. Theoretically, the study contributes to existing PPP governance literature by identifying governance variables specific to Myanmar's institutional and economic context. It emphasizes the importance of tailoring governance frameworks to suit the regulatory maturity, institutional capacity, and financial constraints of developing economies. It also acknowledges the relevance of global studies on citizen participation and financial modeling, particularly in energy system transitions, thereby broadening theoretical perspectives [58].

Practically, the study provides insights for policymakers and project managers to address governance gaps and improve PPP outcomes in energy infrastructure [60]. These findings are applicable not only to Myanmar but also to other developing countries facing similar governance challenges.

4.6 Limitation

While this study provides valuable insights into the governance of PPP projects in Myanmar, several limitations must be acknowledged. Its reliance on case studies may not fully capture the diversity of PPP projects across different regions and sectors. Additionally, Myanmar's rapidly changing political and economic environment could restrict the applicability of these findings. Future research should incorporate a broader range of case studies and consider longitudinal approaches to account for the dynamic nature of PPP governance, potentially extending the findings to other developing countries. Because interview respondents were union-level officials, the perspectives primarily reflect institutional viewpoints; although triangulation with contracts, procurement records, and policy documents mitigates this limitation, broader stakeholder perspectives may further enrich future studies. Section 5 concludes by synthesizing contributions and outlining reform priorities and research avenues.

5 Conclusion

This study examined governance in energy PPPs in a developing-economy context, focusing on Myanmar's union-level institutions. Using four comparative cases in hydropower and oil and gas (two successful, two unsuccessful), we identified 22 governance factors that shape outcomes and clarified the mechanisms through which they operate. The contribution is a practice-tested and transferable framework that links legal–regulatory arrangements, institutional coordination, stakeholder processes, and financial/risk structuring to performance in capital-intensive, long-horizon projects.

Three nuanced insights emerge. First, centralized decision rights must be paired with strong interministerial coordination and reliable data flows. Without effective PPP Unit coordination, shared data standards, and proactive regulators, approval processes become fragmented and project delays compound. Second, a persistent policy–practice disconnect undermines execution; transparency, guideline use, and contract management must be operationalized (e.g., through routinized compliance audits and clearly defined approval pathways) rather than symbolic. Third, financial innovation paired with clear, communicated risk allocation improves bankability and resilience; fit-for-purpose instruments and tax incentives are insufficient unless risk-sharing rules are intelligible and consistently applied. Together, these insights suggest the need for an integrated reform path calibrated to local constraints.

Implications for policy and practice include codifying transparent procurement and contract oversight, empowering the PPP Unit to convene cross-agency data sharing with clear service-level expectations, institutionalizing dispute prevention and resolution routines, adopting a standardized risk-allocation framework with communication protocols, and piloting blended-finance/guarantee options where fiscal space is limited. For practitioners, the framework offers a diagnostic tool to prioritize governance fixes with the highest influence on performance.

Limitations and future research. This multiple-case design relies on a small-N sample drawn from union-level institutions and elite informants, with data collected within a time-bounded 2024 window; the generalizability of the findings beyond comparable central-government settings remains limited. Future research should include longitudinal tracking of governance and project performance, extend comparisons to subnational or provincial PPPs and private-sector counterparties, and employ mixed methods (e.g., structured process-tracing, contract text analytics, and targeted surveys) to test transferability and strengthen causal inference—particularly around risk-allocation communication, coordination remedies, and financing innovation.

In sum, the findings advance PPP governance scholarship for resource-constrained settings and provide actionable guidance for policy design, contract management, and capacity building in energy infrastructure.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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